Backend Development Lab

Programs 1, 2, 3 and 6 — code with clear indentation

1. File Handling — Store and Retrieve Student Records

```
f = open("student.txt", "w")
n = input("Enter name: ")
r = input("Enter roll no: ")
f.write(n + "," + r + "\\n")
f.close()

f = open("student.txt", "r")
print("Student Records:")
print(f.read())
f.close()
```

Remember: `open()`, `write()`, `read()`, `close()`.

2. Classes, Objects, Inheritance and Polymorphism

```
class Employee:
    def display(self):
        print("Employee details")

class Manager(Employee):
    def display(self):
        print("Manager details")

e = Employee()
m = Manager()
e.display()
m.display()
```

Remember: The four spaces before statements inside each method are important. `Manager(Employee)` shows inheritance.

3. Read and Process CSV and JSON Files

```
import csv
import json

print("CSV Records:")
with open("students.csv") as f:
    for row in csv.DictReader(f):
        print(row)

print("\\nJSON Records:")
with open("students.json") as f:
    for student in json.load(f):
        print(student)
```

students.csv

```
id,name,course
1,Rahul,BCA
2,Anu,BCA
```

students.json

```
[
    {"id": 1, "name": "Rahul", "course": "BCA"},
    {"id": 2, "name": "Anu", "course": "BCA"}
]
```

Remember: Keep `data.py`, `students.csv`, and `students.json` in the same folder.

6. MySQL Student Management Database

```
CREATE DATABASE college;
USE college;
```

```
CREATE TABLE Student (
    id INT PRIMARY KEY,
    name VARCHAR(50),
    course VARCHAR(30)
);

CREATE TABLE Marks (
    id INT PRIMARY KEY,
    student_id INT,
    marks INT,
    FOREIGN KEY (student_id) REFERENCES Student(id)
);

INSERT INTO Student VALUES (1, 'Rahul', 'BCA');
INSERT INTO Marks VALUES (1, 1, 85);

SELECT * FROM Student;
SELECT * FROM Marks;
```

Remember: `PRIMARY KEY` uniquely identifies a row; `FOREIGN KEY` connects the tables.